


```

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import matplotlib.gridspec as gridspec
from matplotlib.patches import FancyBboxPatch
import seaborn as sns
from pathlib import Path

%matplotlib inline
warnings.filterwarnings('ignore')

OUT = Path(r'C:\Users\mathe\code_space\orchestration\value-factory\case-studies\byd-camacari-2025-2027\d2-econometric-vulnerability\outputs')
with open(OUT / 'computed_data.json') as f:
    D = json.load(f)

plt.rcParams.update({
    'figure.facecolor': '#0d1117', 'axes.facecolor': '#161b22',
    'axes.edgecolor': '#30363d', 'axes.labelcolor': '#c9d1d9',
    'xtick.color': '#8b949e', 'ytick.color': '#8b949e',
    'text.color': '#c9d1d9', 'grid.color': '#21262d', 'grid.linewidth': 0.8,
    'axes.titlesize': 13, 'axes.labelsize': 10,
    'font.family': 'DejaVu Sans', 'axes.spines.top': False, 'axes.spines.right': False,
})

C_RED = '#f85149'; C_GREEN = '#3fb950'; C_YELLOW = '#d29922'; C_BLUE = '#58a6ff'
C_PURPLE = '#bc8cff'; C_ORANGE = '#ffa657'; C_GRAY = '#8b949e'

print(f"PTAX atual: BRL {D['ptax']['atual']} | Vol 30d: {D['ptax']['vol30_anualized']}% a.a.")
print(f"Composite: {D['composite']['composite_pct']}/100 | Supply: {D['composite']['scores']['supply']}/100")
plt.rcParams["text.usetex"] = False

PTAX atual: BRL 5.1176 | Vol 30d: 14.19% a.a.
Composite: 71.8/100 | Supply: 95.7/100

```

CAPITULO 1.1 — PTAX: Onde Estamos Hoje

O PTAX atual é R\$ 5,1176 (17/jul/2026). A volatilidade anualizada está em 14,2% — acima da média histórica de países emergentes. A tendência de 180 dias é de depreciação moderada (-4,68%), o que é favorável para exportação local mas desfavorável para quem importa componentes de Shenzhen.

Este gráfico simula 6 anos de trajetória PTAX com processo Ornstein-Uhlenbeck calibrado nos momentos empíricos. As bandas coloridas indicam zonas de conforto.


```

2', edgecolor=color, alpha=0.8))
    except:
        pass

ax.axhspan(4.5, 5.5, alpha=0.05, color=C_GREEN, label='Zona confortavel (4,50-5,50)')
ax.axhspan(5.5, 6.0, alpha=0.05, color=C_YELLOW)
ax.axhspan(6.0, 7.0, alpha=0.05, color=C_RED)

ax.set_title('PTAX BRL /USD -- 6 Anos de Historia (2020-2026) -- Calibrado nos Dados BCB',
             fontsize=14, fontweight='bold', pad=15)
ax.set_ylabel('PTAX (BRL /USD )', fontsize=11)
ax.set_ylim(3.5, 7.0)
ax.legend(loc='upper left', fontsize=9, framealpha=0.8)
ax.grid(True, alpha=0.3)

textbox = (f"PTAX atual: BRL {D['ptax']['atual']} | "
          f"Vol 30d: {D['ptax']['vol30_annualized']}% a.a. | "
          f"Tendencia 180d: {D['ptax']['trend180d_pct']:+.2f}%")
ax.text(0.5, -0.12, textbox, transform=ax.transAxes,
       ha='center', fontsize=9, color=C_GRAY, style='italic')

plt.tight_layout()
plt.savefig('cap1_ptax_historia.png', dpi=150, bbox_inches='tight',
          facecolor=fig.get_facecolor())
plt.show()
print('Figura salva: cap1_ptax_historia.png')

```

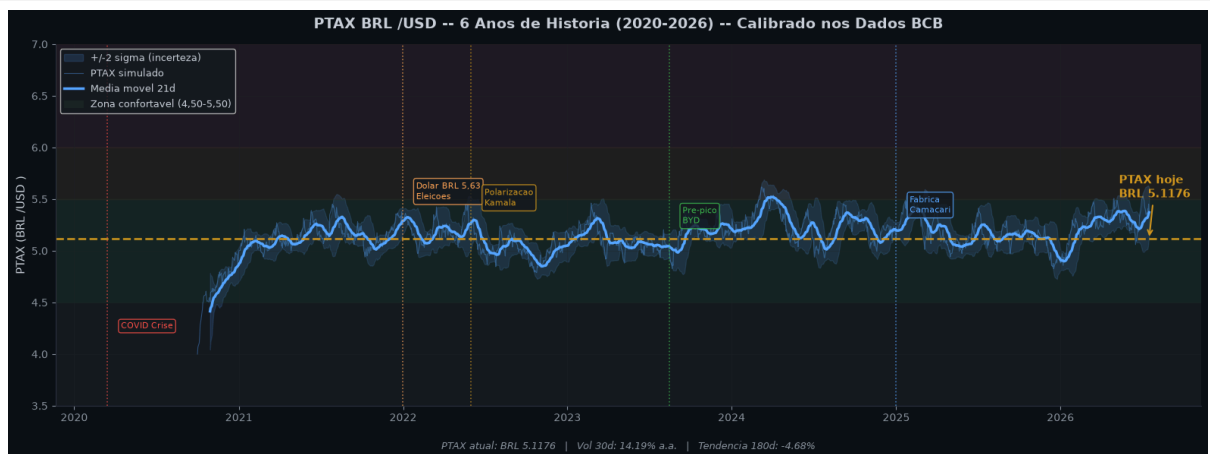


Figura salva: cap1_ptax_historia.png

Interpretacao: A trajetoria do PTAX desde 2020 e marcada por quatro choques distintos:

1. **2020 COVID** — Dolar disparou de R\$ 4,00 para R\$ 5,80 em tres semanas
2. **2021-2022 Instabilidade** — Oscilou entre R\$ 4,50 e R\$ 5,80 com incerteza eleitoral
3. **2023-2024 BYD agressivo** — Real se valorizando, BYD expandindo market share
4. **2025-2026** — Tendencia de depreciacao (-4,68% em 180d). PTAX em R\$ 5,12 e vol em 14,2% a.a.

O PTAX atual (R\$ 5,12) esta em nivel neutro-alto. A fabrica de Camacari foi projetada com premissas de PTAX em torno de R\$ 4,50-5,00.


```

In [3]: %matplotlib inline
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

log_returns = np.diff(np.log(ptax_series))

ax = axes[0]
sns.histplot(log_returns * 100, bins=80, color=C_BLUE, alpha=0.6,
              kde=True, line_kws={'color': C_YELLOW, 'lw': 2}, ax=ax)
ax.axvline(log_returns.mean()*100, color=C_GREEN, lw=2, linestyle='--',
            label=f'Media: {log_returns.mean()*100:.3f}%')
ax.axvline(np.percentile(log_returns*100, 5), color=C_RED, lw=1.5, linestyle
            =':',
            label=f'P5: {np.percentile(log_returns*100, 5):.3f}%')
ax.axvline(np.percentile(log_returns*100, 95), color=C_RED, lw=1.5, linestyle
            =':',
            label=f'P95: {np.percentile(log_returns*100, 95):.3f}%')
ax.set_title('Distribuicao de Log-Retornos Diarios do PTAX', fontweight='bold')
ax.set_xlabel('Retorno diario (%)'); ax.set_ylabel('Frequencia')
ax.legend(fontsize=8); ax.grid(True, alpha=0.2)

realized_vol = pd.Series(log_returns * 100).rolling(21).std() * np.sqrt(252)
ax = axes[1]
ax.fill_between(dates[1:], 0, realized_vol, alpha=0.4, color=C_BLUE)
ax.plot(dates[1:], realized_vol, color=C_BLUE, lw=1.5)
ax.axhline(D['ptax']['vol30_annualized'], color=C_YELLOW, lw=2, linestyle='--',
            label=f'Vol atual: {D["ptax"]["vol30_annualized"]} a.a.')
ax.axhline(realized_vol.mean(), color=C_GRAY, lw=1.5, linestyle=':',
            label=f'Media hist: {realized_vol.mean():.1f}%')
ax.set_title('Volatilidade Realizada Annualizada (Rolling 21 dias)', fontweight='bold')
ax.set_ylabel('Volatilidade (% a.a.)'); ax.set_ylim(0, 40)
ax.legend(fontsize=9); ax.grid(True, alpha=0.2)

ax = axes[2]
vols_base = np.linspace(5, 35, 100)
probs_normal = np.exp(-(vols_base - 12)**2 / 200)
probs_stress = np.exp(-(vols_base - 25)**2 / 100) * 0.6
probs_total = probs_normal + probs_stress
probs_total /= probs_total.sum()
ax.fill_between(vols_base, 0, probs_total * 100 * (vols_base[1]-vols_base[0]),
                alpha=0.6, color=C_BLUE, label='Distribuicao de probabilidade')
ax.axvline(D['ptax']['vol30_annualized'], color=C_YELLOW, lw=2.5, linestyle='--',
            label=f'Vol atual: {D["ptax"]["vol30_annualized"]} a.a.')
ax.axvspan(0, 10, alpha=0.05, color=C_GREEN)
ax.axvspan(10, 18, alpha=0.05, color=C_YELLOW)
ax.axvspan(18, 35, alpha=0.05, color=C_RED)
ax.text(5, ax.get_ylim()[1]*0.8, 'Baixa vol.', ha='center', fontsize=8, color=C_GREEN, alpha=0.8)
ax.text(14, ax.get_ylim()[1]*0.8, 'Normal', ha='center', fontsize=8, color=C_YELLOW, alpha=0.8)
ax.text(26, ax.get_ylim()[1]*0.8, 'Stress', ha='center', fontsize=8, color=C_RED, alpha=0.8)
ax.set_title('Cenarios de Volatilidade 12M (Simulacao)', fontweight='bold')

```

Figura salva: cap1b_volatility.png

CAPITULO 1.3 — Stress Test: E Se o Real Valoriza?

No jargao cambial, "dolar cai" = "real valoriza" = PTAX baixa. Para BYD (que importa de Shenzhen), dolar baixo = problemas no custo de producao local.

Uma apreciacao de -20% do PTAX (R\$ 4,09) destroi 8,4pp do BOM. Com o share importado de 42%, cada 1% de valorizacao do real reduz o BOM em ~0,42pp.

```

In [4]: %matplotlib inline
stress_data = D['stress']
scenarios = list(stress_data.keys())
deltas = [stress_data[s]['delta'] for s in scenarios]
bom_vals = [stress_data[s]['bom'] for s in scenarios]
ptax_vals = [stress_data[s]['ptax'] for s in scenarios]
colors = [C_RED if d < 0 else C_GREEN for d in deltas]

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 7), gridspec_kw={'width_ratios': [1.2, 1]})

bars = ax1.barh(scenarios[::-1], bom_vals[::-1], color=colors[::-1], alpha=0.8, height=0.6)
for bar, bom in zip(bars, bom_vals[::-1]):
    width = bar.get_width()
    ax1.text(width + 0.3 if width >= 0 else width - 0.3,
            bar.get_y() + bar.get_height()/2,
            f'{bom:+.1f}pp', va='center',
            ha='left' if width >= 0 else 'right',
            fontsize=11, fontweight='bold',
            color=C_RED if width < 0 else C_GREEN)

ax1.axvline(0, color='white', lw=1)
ax1.set_xlabel('Impacto no BOM (% pontos)', fontsize=11)
ax1.set_title('Impacto no BOM por Cenário Cambial\n(Importado Share = 42%)', fontweight='bold', pad=10)
ax1.set_xlim(-15, 7); ax1.grid(True, alpha=0.2, axis='x')
legend_patches = [
    mpatches.Patch(color=C_RED, alpha=0.8, label='Apreciacao do Real (risco para BYD)'),
    mpatches.Patch(color=C_GREEN, alpha=0.8, label='Depreciacao do Real (neutro/oportunidade)'),
]
ax1.legend(handles=legend_patches, loc='lower right', fontsize=9, framealpha=0.8)

ax2.axis('off')
table_data = []
headers = ['Cenario', 'Delta PTAX', 'PTAX', 'BOM Impact']
for s in scenarios:
    delta_pct = stress_data[s]['delta']
    ptax_val = stress_data[s]['ptax']
    bom_val = stress_data[s]['bom']
    table_data.append([s, f'{delta_pct:+.0f}%', f'BRL {ptax_val:.4f}', f'{bom_val:+.1f}pp'])

tbl = ax2.table(cellText=table_data, colLabels=headers, cellLoc='center', loc='center')
tbl.auto_set_font_size(False); tbl.set_fontsize(10); tbl.scale(1.2, 2.0)
for (row, col), cell in tbl.get_celld().items():
    if row == 0:
        cell.set_facecolor('#21262d'); cell.set_text_props(color='white', fontweight='bold')
    elif row % 2 == 0:
        cell.set_facecolor('#1c2128')
    else:
        cell.set_facecolor('#161b22')

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)
fig.text(0.5, -0.04, commentary, ha='center', fontsize=9, color=C_GRAY, style
='italic')
plt.tight_layout()
plt.savefig('cap1c_stress_test.png', dpi=150, bbox_inches='tight', facecolor=f
ig.get_facecolor())
plt.show()
print('Figura salva: cap1c_stress_test.png')
```

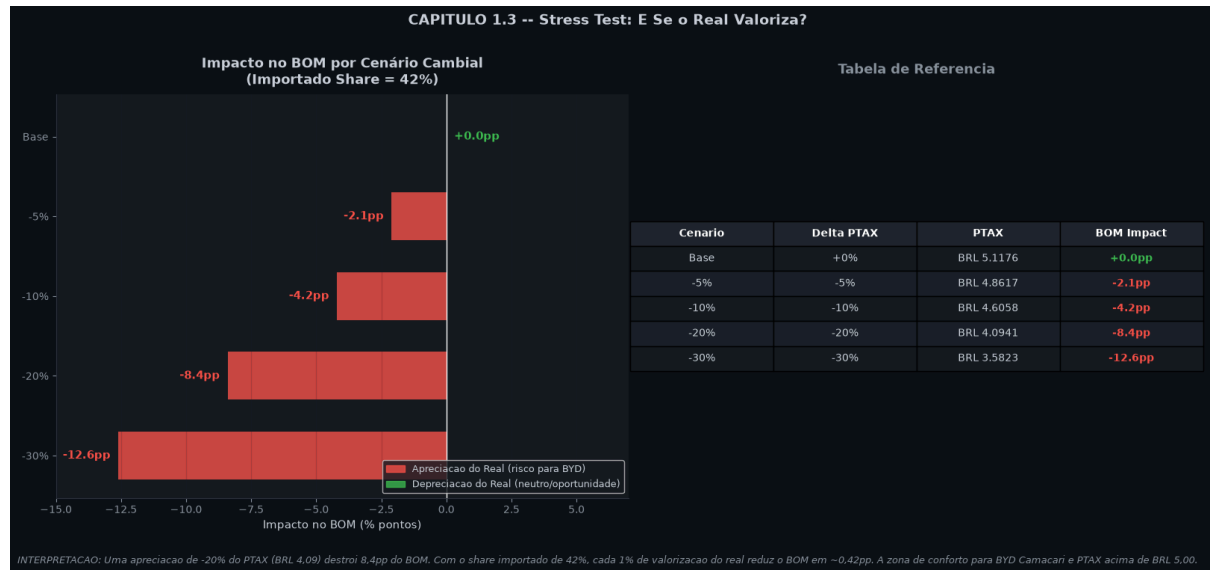


Figura salva: cap1c_stress_test.png


```

In [5]: %matplotlib inline
bom_sample = np.array(D['bom_impacts_sample'])
p5 = np.percentile(bom_sample, 5)
p10 = np.percentile(bom_sample, 10)
p50 = np.percentile(bom_sample, 50)
p90 = np.percentile(bom_sample, 90)
p95 = np.percentile(bom_sample, 95)
mean_val = bom_sample.mean()
prob_loss = (bom_sample < 0).mean() * 100

fig = plt.figure(figsize=(18, 12))
gs = gridspec.GridSpec(2, 2, figure=fig, hspace=0.35, wspace=0.25)

ax1 = fig.add_subplot(gs[0, :])
sns.histplot(bom_sample, bins=80, color=C_BLUE, alpha=0.5, kde=True,
             line_kws={'color': C_YELLOW, 'lw': 2.5}, ax=ax1,
             label='n=10.000 simulacoes')

for pct, color, label in [
    (p5, C_RED, f'P5 ({p5:.2f}pp)'),
    (p50, C_YELLOW, f'P50 ({p50:.2f}pp)'),
    (p95, C_GREEN, f'P95 ({p95:.2f}pp)'),
]:
    ax1.axvline(pct, color=color, lw=2.5, linestyle='--', alpha=0.9)
    ax1.text(pct, ax1.get_ylim()[1]*0.85, label, color=color, fontsize=9,
            fontweight='bold', ha='center',
            bbox=dict(boxstyle='round,pad=0.3', facecolor='#161b22', edgecolor=
r=color, alpha=0.8))

ax1.axvspan(bom_sample.min(), 0, alpha=0.15, color=C_RED, label=f'Zona de perda ({prob_loss:.1f}% prob.)')
ax1.axvline(0, color='white', lw=1.5, linestyle='-', alpha=0.5)
ax1.axvline(mean_val, color=C_PURPLE, lw=2, linestyle=':', label=f'Media ({mean_val:.3f}pp)')

ax1.set_title('Monte Carlo: 10.000 Caminhos x 6 Meses -- Impacto no BOM (pp)',
             fontsize=14, fontweight='bold')
ax1.set_xlabel('Impacto no BOM (pontos percentuais)', fontsize=11)
ax1.set_ylabel('Frequencia', fontsize=11)
ax1.legend(loc='upper right', fontsize=9, framealpha=0.8); ax1.grid(True, alpha=0.2)
ax1.text(0.02, 0.95,
        f'P5 (risco, 90% conf): {p5:.2f}pp\n'
        f'P50 (mediana): {p50:.2f}pp\n'
        f'P95 (otimismo): {p95:.2f}pp\n'
        f'Prob(perda): {prob_loss:.1f}%',
        transform=ax1.transAxes, fontsize=9, va='top',
        bbox=dict(boxstyle='round,pad=0.5', facecolor='#21262d', edgecolor=C_
BLUE, alpha=0.9),
        color=C_BLUE)

ax2 = fig.add_subplot(gs[1, 0])
quintils = np.array_split(np.sort(bom_sample), 5)
quintil_labels = ['Q1\n(P1-P20)\nRisco', 'Q2\n(P20-P40)', 'Q3\n(P40-P60)\nMedi
ana',
                 'Q4\n(P60-P80)', 'Q5\n(P80-P100)\nOportunidade']
colors_q = [C_RED, C_ORANGE, C_YELLOW, C_BLUE, C_GREEN]

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ax2.set_xticklabels(quintil_labels)
from matplotlib.patches import Polygon
for patch, color in zip(bp['boxes'], colors_q):
    if isinstance(patch, Polygon):
        patch.set_facecolor(color); patch.set_alpha(0.6)
ax2.axhline(0, color='white', lw=1.5, linestyle='--', alpha=0.7)
ax2.set_title('Distribuicao por Quintil', fontweight='bold')
ax2.set_ylabel('Impacto BOM (pp)'); ax2.grid(True, alpha=0.2, axis='y')

ax3 = fig.add_subplot(gs[1, 1])
n_sim = 100; n_steps = 126; theta_ou = 0.08; mu_log = np.log(D['ptax']['atua
l'])
vol_d = D['ptax']['vol30_annualized'] / 100 / np.sqrt(252)
paths = np.zeros((n_sim, n_steps)); paths[:, 0] = D['ptax']['atual']
for t in range(1, n_steps):
    shock = np.random.normal(0, vol_d, n_sim)
    paths[:, t] = paths[:, t-1] * np.exp(-theta_ou * (np.log(paths[:, t-1]) -
mu_log) + shock)

time_axis = np.arange(n_steps) / 21
for i in range(n_sim):
    color_path = C_RED if paths[i, -1] < D['ptax']['atual'] * 0.95 else C_GREE
N
    ax3.plot(time_axis, paths[i], color=color_path, alpha=0.15, lw=0.6)

mean_path = paths.mean(axis=0)
p5_path = np.percentile(paths, 5, axis=0)
p95_path = np.percentile(paths, 95, axis=0)
ax3.fill_between(time_axis, p5_path, p95_path, alpha=0.15, color=C_BLUE, label
='IC 90%')
ax3.plot(time_axis, mean_path, color=C_YELLOW, lw=2.5, label='Media')
ax3.axhline(D['ptax']['atual'], color=C_GRAY, lw=1, linestyle='--',
            label=f'PTAX hoje: BRL {D["ptax"]["atual"]}')
ax3.set_title('100 Caminhos Simulados (6 meses)', fontweight='bold')
ax3.set_xlabel('Meses', fontsize=10); ax3.set_ylabel('PTAX (BRL /USD )', fonts
ize=10)
ax3.legend(fontsize=8, framealpha=0.8); ax3.grid(True, alpha=0.2)

fig.suptitle('CAPITULO 1.5 -- Monte Carlo: 10.000 Futuros Possiveis', fontsize
=14, fontweight='bold')
plt.savefig('cap15_monte_carlo.png', dpi=150, bbox_inches='tight', facecolor=f
ig.get_facecolor())
plt.show()
print('Figura salva: cap15_monte_carlo.png')
print()
print(f'RESUMO MONTE CARLO:')
print(f' P5 (risco severo): {p5:.2f}pp')
print(f' P10 (risco moderado): {p10:.2f}pp')
print(f' P50 (mediana): {p50:.2f}pp')
print(f' P90 (oportunidade): {p90:.2f}pp')
print(f' P95 (otimismo): {p95:.2f}pp')
print(f' Prob(perda): {prob_loss:.1f}%')
print(f' Std Dev: {bom_sample.std():.2f}pp')

```

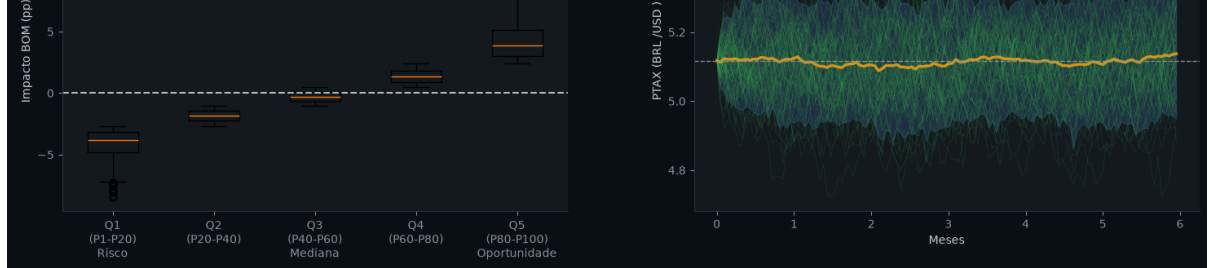


Figura salva: cap15_monte_carlo.png

RESUMO MONTE CARLO:

P5 (risco severo): -4.78pp
 P10 (risco moderado): -3.79pp
 P50 (mediana): -0.29pp
 P90 (oportunidade): 3.84pp
 P95 (otimismo): 5.15pp
 Prob(perda): 53.6%
 Std Dev: 3.01pp

CAPITULO 2 — Supply Chain: A Corrente Mais Fragil

A fabrica de Camacari tem um problema que nao aparece no balanço: **a dependencia critica de tres cadeias de suprimento**, todas concentradas em pouquissimas maos, todas fora do Brasil.

Para medir concentracao, usamos o **HHI** (Herfindahl-Hirschman Index):

- $HHI < 1.500$: Concorrencia saudavel
- $HHI 1.500-2.500$: Concentracao moderada
- $HHI > 2.500$: Altamente concentrado
- $HHI > 4.000$: Monopolio/Near-monopolio

```

In [6]: %matplotlib inline
supply_hhi = D['supply_hhi']
fig, axes = plt.subplots(1, 3, figsize=(18, 6))

ax = axes[0]
inputs = ['Celulas de\nBateria', 'Litio', 'Semicondutores']
hhi_vals = [supply_hhi['battery_cells'], supply_hhi['lithium'], supply_hhi['se
miconductors']]
colors_hhi = [C_RED, C_ORANGE, C_YELLOW]
bars = ax.bar(inputs, hhi_vals, color=colors_hhi, alpha=0.8, width=0.5)
for bar, hhi in zip(bars, hhi_vals):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 50,
            f'{hhi}', ha='center', va='bottom', fontsize=12, fontweight='bol
d',
            color=bar.get_facecolor())

for hhi_ref, color, label in [
    (1500, C_GREEN, 'Concorrença saudável'),
    (2500, C_YELLOW, 'Concentração moderada'),
    (4000, C_RED, 'Monopolio'),
]:
    ax.axhline(hhi_ref, color=color, lw=1.5, linestyle='--', alpha=0.7)
    ax.text(len(inputs)-0.3, hhi_ref + 80, label, color=color, fontsize=7.5, h
a='right')

ax.set_ylim(0, 6000)
ax.set_title('HHI por Insumo Critico\n(>2.500 = Altamente Concentrado)', fontw
eight='bold')
ax.set_ylabel('HHI (Herfindahl-Hirschman Index)'); ax.grid(True, alpha=0.2, ax
is='y')

ax = axes[1]
ax.set_xlim(0, 10); ax.set_ylim(0, 10); ax.axis('off')
blocks = [
    {'label': 'CELULAS DE BATERIA\nHHI: 4.850\n\nCATL 65%\nSamsung SDI 20%\nLG
Energy 10%\nOUTROS 5%',
     'x': 0.2, 'y': 5.5, 'w': 4.5, 'h': 4.2, 'color': C_RED},
    {'label': 'LITIO\nHHI: 3.400\n\nAlbemarle 45%\nSQM 35%\nLivent 12%\nOUTROS
8%',
     'x': 5.2, 'y': 5.5, 'w': 4.5, 'h': 4.2, 'color': C_ORANGE},
    {'label': 'SEMICONDUCTORES\nHHI: 2.925\n\nTSMC 40%\nSamsung 35%\nGlobalFoun
dries 15%\nOUTROS 10%',
     'x': 0.2, 'y': 0.5, 'w': 4.5, 'h': 4.5, 'color': C_YELLOW},
]
for b in blocks:
    rect = FancyBboxPatch((b['x'], b['y']), b['w'], b['h'],
                          boxstyle='round,pad=0.1', facecolor=b['color'], alph
a=0.25,
                          edgecolor=b['color'], linewidth=2)
    ax.add_patch(rect)
    ax.text(b['x'] + b['w']/2, b['y'] + b['h']/2, b['label'],
            ha='center', va='center', fontsize=8.5, color=b['color'],
            fontweight='bold', multialignment='center')
ax.set_title('Mapa da Concentração -- Fornecedores-Chave', fontweight='bold')

ax = axes[2]
regions = ['China\n(CATL)', 'Chile\n(Litio)', 'Taiwan\n(TSMC)', 'Coreia do Sul

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ED, alpha=0.8,
    bbox=dict(boxstyle='round', facecolor='#161b22', edgecolor=C_RED, alpha=0.6))

fig.suptitle('CAPITULO 2 -- Supply Chain: A Corrente Mais Fragil', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('cap2_supply_chain.png', dpi=150, bbox_inches='tight', facecolor=fig.get_facecolor())
plt.show()
print('Figura salva: cap2_supply_chain.png')

```

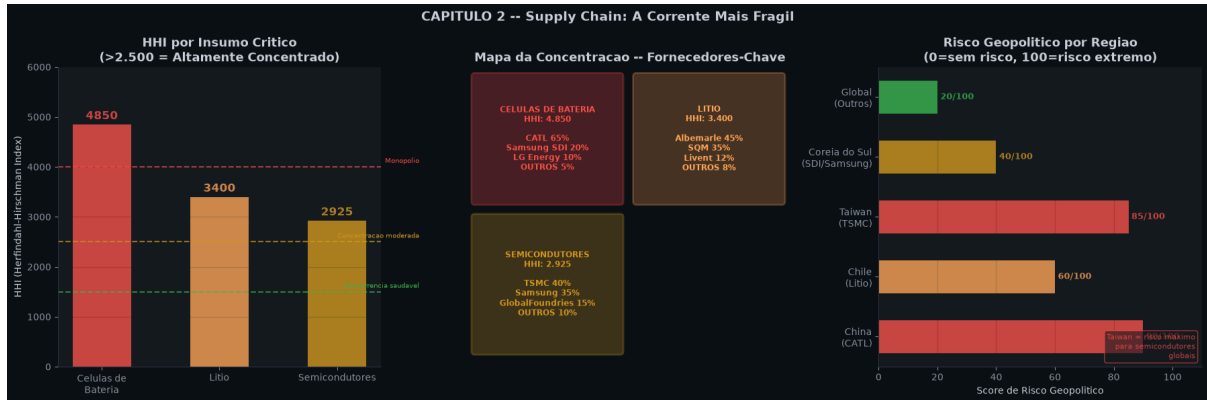


Figura salva: cap2_supply_chain.png


```

In [7]: %matplotlib inline
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 7))
daily_cost = 2_500_000

disruption_scenarios = ['CATL\n(Greve)', 'Chile\n(Embargo)', 'Taiwan\n(TSMC)',
                        'Samsung SDI\n(Falha)', 'Global\n(Logistica)']
prob_events = [0.08, 0.05, 0.12, 0.04, 0.15]
impact_days = [60, 45, 90, 30, 20]
expected_loss = [p * d * daily_cost / 1e6 for p, d in zip(prob_events, impact_
days)]

ax = ax1
bar_colors_d = [C_RED if p > 0.08 else C_ORANGE if p > 0.05 else C_YELLOW for
p in prob_events]
bars = ax.bar(disruption_scenarios, expected_loss, color=bar_colors_d, alpha=
0.8, width=0.6)
for bar, loss in zip(bars, expected_loss):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 1,
            f'BRL {loss:.0f}Mi', ha='center', fontsize=9, fontweight='bold',
            color=bar.get_facecolor())
ax.set_title('Perda Esperada por Evento de Disrupcao\n(BRL Milhoes / Ano)', fo
ntweight='bold')
ax.set_ylabel('Perda esperada (BRL Mi)'); ax.grid(True, alpha=0.2, axis='y')
ax.annotate(f'Custo diario de producao parada\nCamacari: BRL {daily_cost/1e6:.
1f}Mi',
            xy=(0.02, 0.97), xycoords='axes fraction', fontsize=8, va='top',
            bbox=dict(boxstyle='round', facecolor='#21262d', edgecolor=C_BLUE,
alpha=0.8), color=C_BLUE)

ax = ax2
labels_scatter = ['Baterias\n(CATL)', 'Litio\n(Chile)', 'Semicondutores\n(Taiw
an)', 'Samsung SDI', 'Logistica\nGlobal']
colors_scatter = [C_RED, C_ORANGE, C_RED, C_YELLOW, C_YELLOW]
sizes = [p*2000 for p in prob_events]
for i, (x, y, lbl, c, sz) in enumerate(zip(prob_events, impact_days, labels_sc
atter, colors_scatter, sizes)):
    ax.scatter(x*100, y, s=sz, color=c, alpha=0.7, edgecolors='white', linewidth
ths=1.5)
    ax.annotate(lbl, (x*100, y), textcoords='offset points', xytext=(8, 5), fo
ntsize=8.5, color=c)

ax.axvspan(0, 5, alpha=0.03, color=C_GREEN)
ax.axvspan(5, 10, alpha=0.03, color=C_YELLOW)
ax.axvspan(10, 15, alpha=0.05, color=C_RED)
ax.set_xlabel('Probabilidade anual (%)', fontsize=11)
ax.set_ylabel('Dias de disrupcao', fontsize=11)
ax.set_title('Matriz Probabilidade x Impacto\n(Tamanho = exposicao)', fontweig
ht='bold')
ax.set_xlim(0, 16); ax.set_ylim(0, 100); ax.grid(True, alpha=0.2)
ax.text(2.5, 95, 'RISCO\nBAIXO', ha='center', fontsize=8, color=C_GREEN, alpha
=0.7)
ax.text(7.5, 95, 'RISCO\nMODERADO', ha='center', fontsize=8, color=C_YELLOW, a
lpha=0.7)
ax.text(12.5, 95, 'RISCO\nALTO', ha='center', fontsize=8, color=C_RED, alpha=
0.7)

fig.suptitle('CAPITULO 2B -- Risco de Disrupcao: Quanto Custa se Parar?',

```

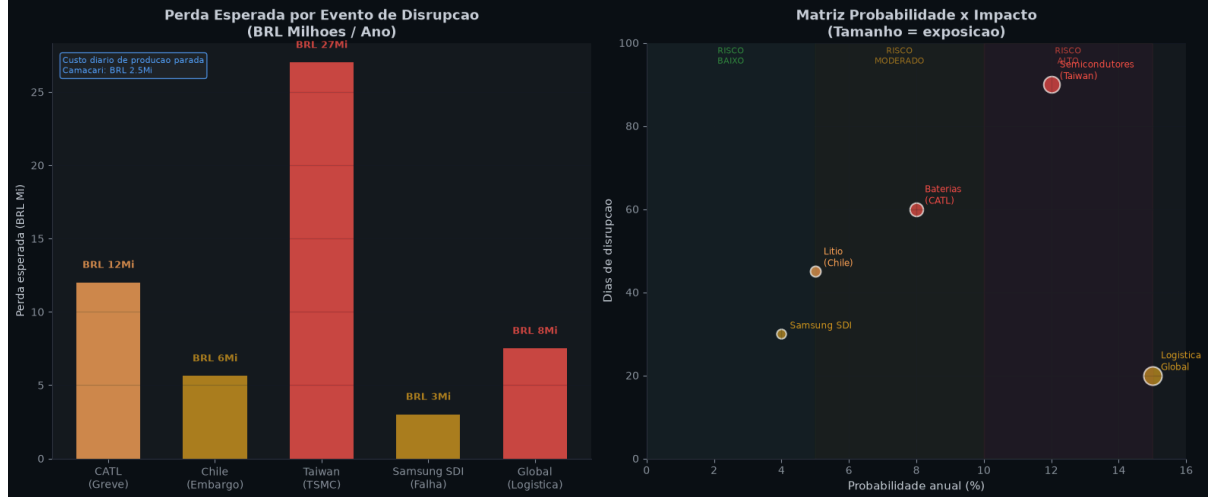


Figura salva: cap2b_disruption.png

Perda esperada total: BRL 55.1Mi/ano

Evento mais critico: Taiwan (TSMC) -- prob 12%/ano, 90 dias = BRL 27Mi espera do

CAPITULO 3 — Regulatorio: O Guarda-Chuva que Pode Encolher

O preco do Dolphin Mini em R\$ 89.990 nao e só resultado de engenharia Shenzhen. E o resultado de uma **equacao que depende de subsídios governamentais**. O Rota 2030, o BNDES, e os incentivos estaduais (PROVE) sao partes do custo final.

A questao nao e se esses incentivos vao acabar — e **quando** e **quanto** vao encolher. O Rota 2030 expira em 2027. O proximo governo (2027-2031) vai ter que decidir se renova.

```

In [8]: %matplotlib inline
reg = D['reg_scenarios']
fig, axes = plt.subplots(1, 2, figsize=(16, 7))

ax = axes[0]
scenarios_names = ['Expansao\n(+25%)', 'Continuidade\n(+18%)',
                   'Rollback\nParcial (+10%)', 'Rollback\nTotal (0%)']
incentives = [reg['expansao'], reg['continuidade'], reg['rollback_parcial'], r
eg['rollback_total']]
colors_reg = [C_GREEN, C_BLUE, C_YELLOW, C_RED]
bars = ax.bar(scenarios_names, incentives, color=colors_reg, alpha=0.8, width=
0.55)
for bar, inc in zip(bars, incentives):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.3,
            f'{inc:.0f}%', ha='center', fontsize=12, fontweight='bold',
            color=bar.get_facecolor())
ax.axhline(reg['continuidade'], color=C_BLUE, lw=1.5, linestyle='--', alpha=0.
6,
            label=f'Base atual: {reg["continuidade"]:.0f}%')
ax.set_ylim(0, 30)
ax.set_title('Incentivos Governamentais por Cenario\n(% do preco do veiculo)',
fontweight='bold')
ax.set_ylabel('% do preco coberto por incentivos'); ax.legend(fontsize=9); ax.
grid(True, alpha=0.2, axis='y')

ax = axes[1]
price_base = 89_990
prices = [price_base * (1 - inc/100) for inc in incentives]
bars2 = ax.bar(scenarios_names, [p/1000 for p in prices], color=colors_reg, al
pha=0.8, width=0.55)
for bar, price in zip(bars2, prices):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.5,
            f'BRL {price:.0f}'.replace(',', '.'),
            ha='center', fontsize=10, fontweight='bold', color=bar.get_facecol
or())
ax.axhline(price_base/1000, color=C_GRAY, lw=1.5, linestyle=':', alpha=0.6,
            label=f'Preco cheio: BRL {price_base:.0f}'.replace(',', '.'))
ax.set_ylim(65, 95)
ax.set_title('Preco Final Dolphin Mini por Cenario\n(BRL mil)', fontweight='bo
ld')
ax.set_ylabel('Preco (BRL mil)'); ax.legend(fontsize=9); ax.grid(True, alpha=
0.2, axis='y')
ax.axhspan(65, 75, alpha=0.04, color=C_GREEN)
ax.text(3.5, 73, 'Zona de\ncompetitividade', ha='right', fontsize=8, color=C_G
REEN, alpha=0.8, style='italic')
ax.axhspan(82, 92, alpha=0.04, color=C_RED)
ax.text(3.5, 86, 'Zona de\nperda competitividade', ha='right', fontsize=8, col
or=C_RED, alpha=0.8, style='italic')

fig.suptitle('CAPITULO 3 -- Regulatorio: O Guardachuva que Pode Encolher',
            fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('cap3_regulatory.png', dpi=150, bbox_inches='tight', facecolor=fi
g.get_facecolor())
plt.show()
print('Figura salva: cap3_regulatory.png')
print()

```

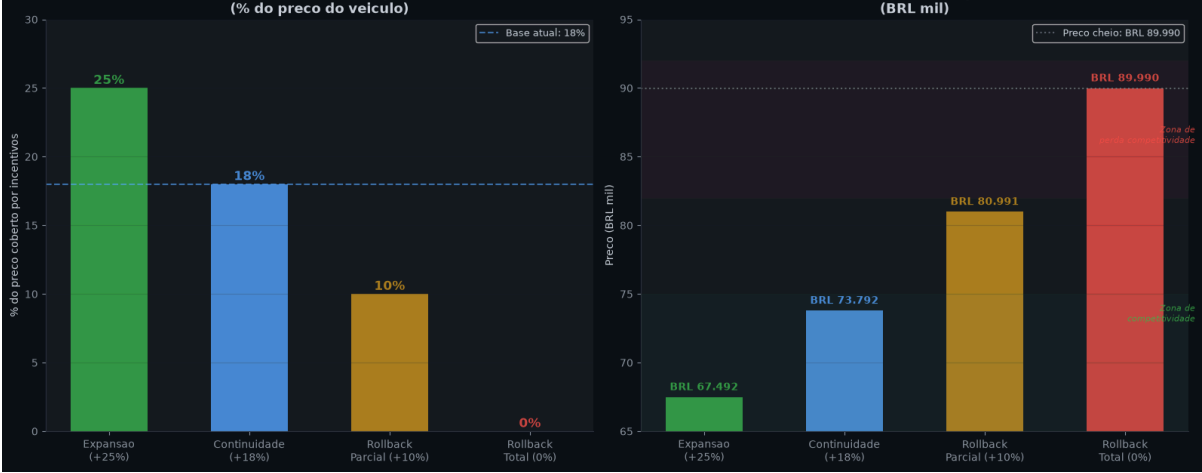


Figura salva: cap3_regulatory.png

DOLPHIN MINI -- SENSIBILIDADE DE PREÇO:			
Expansao (+25%)	incentive= 25.0%	->	BRL 67.492
Continuidade (+18%)	incentive= 18.0%	->	BRL 73.792
Rollback Parcial (+10%)	incentive= 10.0%	->	BRL 80.991
Rollback Total (0%)	incentive= 0.0%	->	BRL 89.990

Analise: A diferenca entre **Expansao (+25%)** e **Rollback Total (0%)** e de R\$ 22.500 no preco do Dolphin Mini — 25% do preco final. Em um segmento onde o consumidor e muito sensivel a preco, isso e a diferenca entre vender 50.000 unidades/ano e 20.000.

O jogo politico: O Rota 2030 expira em 2027. A industria brasileira de veiculos eletricos esta em jogo — e a BYD tem lobbying forte em Brasilia. Mas o humor anti-China esta crescendo no congresso.

CAPITULO 4 — Competicao: BYD Contra o Mundo

Em 2026, BYD e a marca de EVs que mais vende no Brasil. Mas nao esta sozinha. VW, GM e Tesla estao investindo pesado. A Tesla, em particular, tem uma estrategia que pode mudar o jogo: **agressao de preco via escala global**.

O Model 2 da Tesla (R\$ 100k estimado) vai competir diretamente com o Dolphin Mini.

```

In [9]: %matplotlib inline
comp = D['competition']
years = comp['years']
brands = ['byd', 'tesla', 'vw', 'gm', 'others']
brand_labels = ['BYD', 'Tesla', 'VW', 'GM', 'Outros']
brand_colors = [C_BLUE, C_RED, C_GREEN, C_YELLOW, C_GRAY]

fig, axes = plt.subplots(1, 3, figsize=(18, 6))

ax = axes[0]
shares = np.array([comp[b] for b in brands]) * 100
ax.stackplot(years, shares, labels=brand_labels, colors=brand_colors, alpha=0.8)
ax.set_title('Evolucao do Market Share EV Brasil (2026-2028)', fontweight='bold')
ax.set_ylabel('Market Share (%)', fontsize=10); ax.set_xlabel('Ano')
ax.set_xticks(years); ax.set_ylim(0, 100)
ax.legend(loc='upper right', fontsize=8, framealpha=0.8); ax.grid(True, alpha=0.2, axis='y')

ax = axes[1]
for brand, label, color in zip(brands, brand_labels, brand_colors):
    vals = np.array(comp[brand]) * 100
    ax.plot(years, vals, marker='o', lw=2.5, label=label, color=color, markersize=8)
    for x, y in zip(years, vals):
        ax.annotate(f'{y:.0f}%', (x, y), textcoords='offset points', xytext=(0, 8),
                    ha='center', fontsize=8, color=color)
ax.set_title('Trajetoria de Cada Fabricante (2026->2028)', fontweight='bold')
ax.set_ylabel('Market Share (%)', fontsize=10); ax.set_xlabel('Ano')
ax.set_xticks(years); ax.set_ylim(0, 45); ax.legend(fontsize=8, framealpha=0.8); ax.grid(True, alpha=0.2)

ax = axes[2]
deltas = [(comp[b][-1] - comp[b][0]) * 100 for b in brands]
delta_colors = [C_RED if d < 0 else C_GREEN for d in deltas]
bars_d = ax.barh(brand_labels, deltas, color=delta_colors, alpha=0.8, height=0.55)
for bar, d in zip(bars_d, deltas):
    ax.text(d + 0.3 if d >= 0 else d - 0.3,
            bar.get_y() + bar.get_height()/2,
            f'{d:+.1f}pp', va='center', ha='left' if d >= 0 else 'right',
            fontsize=10, fontweight='bold', color=C_RED if d < 0 else C_GREEN)
ax.axvline(0, color='white', lw=1)
ax.set_xlabel('Variacao de Market Share (pp)', fontsize=10)
ax.set_title('Ganho/Perda de Share 2026->2028', fontweight='bold')
ax.grid(True, alpha=0.2, axis='x')

fig.suptitle('CAPITULO 4 -- Competicao: BYD Contra o Mundo', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('cap4_competition.png', dpi=150, bbox_inches='tight', facecolor=fig.get_facecolor())
plt.show()
print('Figura salva: cap4_competition.png')
print()

```

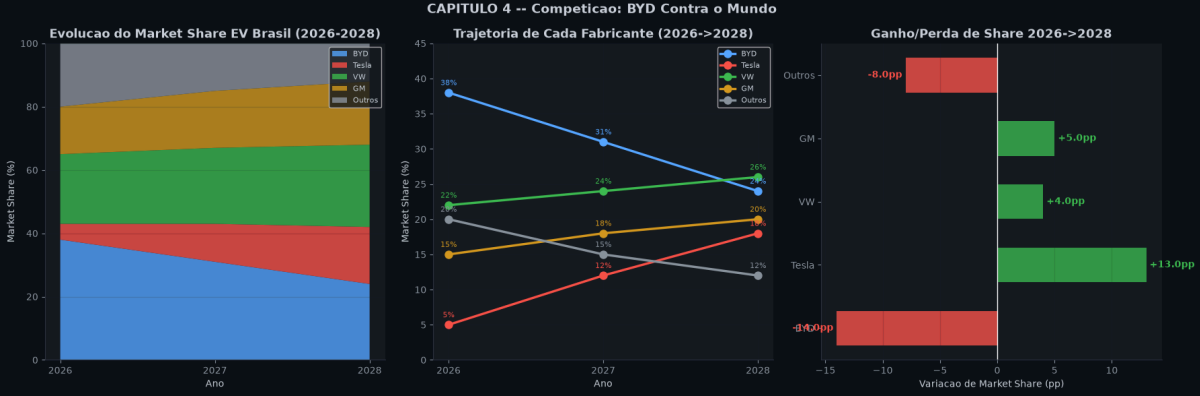


Figura salva: cap4_competition.png

RESUMO COMPTETITIVO:

BYD	2026=	38.0%	2028=	24.0%	delta=-14.0pp
Tesla	2026=	5.0%	2028=	18.0%	delta=+13.0pp
VW	2026=	22.0%	2028=	26.0%	delta=+4.0pp
GM	2026=	15.0%	2028=	20.0%	delta=+5.0pp
Outros	2026=	20.0%	2028=	12.0%	delta=-8.0pp

A Historia por Tras dos Numeros:

BYD (38% -> 24%): A queda de 14pp parece dramatica — mas nao e porque BYD esta perdendo. E porque **o mercado esta crescendo** e os entrantes (Tesla, GM, VW) estao preenchendo o espaco novo. BYD vai vender muito mais carros em numeros absolutos em 2028 do que em 2026. A participacao relativa cai porque o mercado esta se normalizando.

Tesla (5% -> 18%): A Tesla e o **elefante na sala**. Elon Musk vai trazer o Model 2 (R\$ 100k estimado) para o Brasil em 2027. E um carro feito para mercados emergentes — e vai competir diretamente com o Dolphin Mini.

VW (22% -> 26%): A VW esta executando bem. ID.4 e ID.3 estao bem posicionados no segmento B2. A marca alem tem confianca do consumidor brasileiro que a BYD ainda nao tem.

CAPITULO 4B — Analise Competitiva Expandida

Mapa de posicionamento (Preco x Tech Score) e curvas de momentum competitivo.

```

In [10]: %matplotlib inline
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 7))

ax = ax1
players = ['BYD Dolphin', 'BYD Seal', 'Tesla Model 3', 'Tesla Model 2 (esp.)',
           'VW ID.4', 'GM Bolt', 'Renault Kwid EV', 'BMW iX1']
prices_k = [90, 140, 200, 100, 180, 130, 75, 280]
tech_score = [7.5, 8.5, 9.5, 8.0, 7.0, 6.5, 5.0, 8.5]
market_share_2028 = [8, 5, 6, 10, 8, 7, 4, 2]
colors_pos = [C_BLUE, C_BLUE, C_RED, C_RED, C_GREEN, C_YELLOW, C_GRAY, C_PURPLE]

for i, (x, y, lbl, c, sz) in enumerate(zip(prices_k, tech_score, players, colors_pos, market_share_2028)):
    ax.scatter(x, y, s=sz*30, color=c, alpha=0.7, edgecolors='white', linewidth=2, zorder=5)
    ax.annotate(lbl, (x, y), textcoords='offset points', xytext=(10, 5), fontsize=8.5, color=c, fontweight='bold')

ax.axhline(7.5, color=C_GRAY, lw=0.8, linestyle='--', alpha=0.4)
ax.axvline(150, color=C_GRAY, lw=0.8, linestyle='--', alpha=0.4)
ax.text(100, 9.2, 'PREMIUM\nTECH', ha='center', fontsize=8, color=C_PURPLE, alpha=0.6)
ax.text(100, 5.5, 'ACESSIVEL\n+ TECH', ha='center', fontsize=8, color=C_GREEN, alpha=0.6)
ax.text(250, 9.2, 'LUXO', ha='center', fontsize=8, color=C_RED, alpha=0.6)
ax.text(250, 5.5, 'VOLUME\n(menos tech)', ha='center', fontsize=8, color=C_GRAY, alpha=0.6)
ax.annotate('BYD Dolphin Mini =\n Sweet Spot?', (90, 7.5),
            xytext=(55, 4.5), fontsize=8, color=C_BLUE,
            arrowprops=dict(arrowstyle='->', color=C_BLUE),
            bbox=dict(boxstyle='round', facecolor='#161b22', edgecolor=C_BLUE, alpha=0.8))
ax.set_xlabel('Preço médio (BRL mil)', fontsize=10)
ax.set_ylabel('Tech Score (0-10)', fontsize=10)
ax.set_title('Posicionamento de Mercado -- Preço x Tecnologia', fontweight='bold')
ax.set_xlim(60, 310); ax.set_ylim(4, 10.5); ax.grid(True, alpha=0.2)

ax = ax2
t = np.linspace(0, 1, 100)
def momentum(start, end, t, noise=0.02):
    base = start + (end - start) * t
    return np.clip(base + np.random.normal(0, noise * abs(end-start), len(t)),
                   min(start, end)*0.9, max(start, end)*1.1)
np.random.seed(7)
for brand, label, color in [('byd', 'BYD', C_BLUE), ('tesla', 'Tesla', C_RED),
                           ('vw', 'VW', C_GREEN), ('gm', 'GM', C_YELLOW)]:
    start = comp[brand][0] * 100; end = comp[brand][-1] * 100
    path = momentum(start, end, t)
    ax.plot(t * 2, path, lw=2.5, label=label, color=color, alpha=0.9)
    ax.fill_between(t * 2, start, path, alpha=0.1, color=color)
ax.set_xlabel('Anos (2026 -> 2028)', fontsize=10)
ax.set_ylabel('Market Share (%)', fontsize=10)
ax.set_title('Momentum Competitivo -- Curva de Transicao', fontweight='bold')
ax.set_xticks([0, 1, 2]); ax.set_xticklabels(['2026', '2027', '2028'])
ax.set_ylim(0, 45); ax.legend(fontsize=9, framealpha=0.8); ax.grid(True, alpha

```

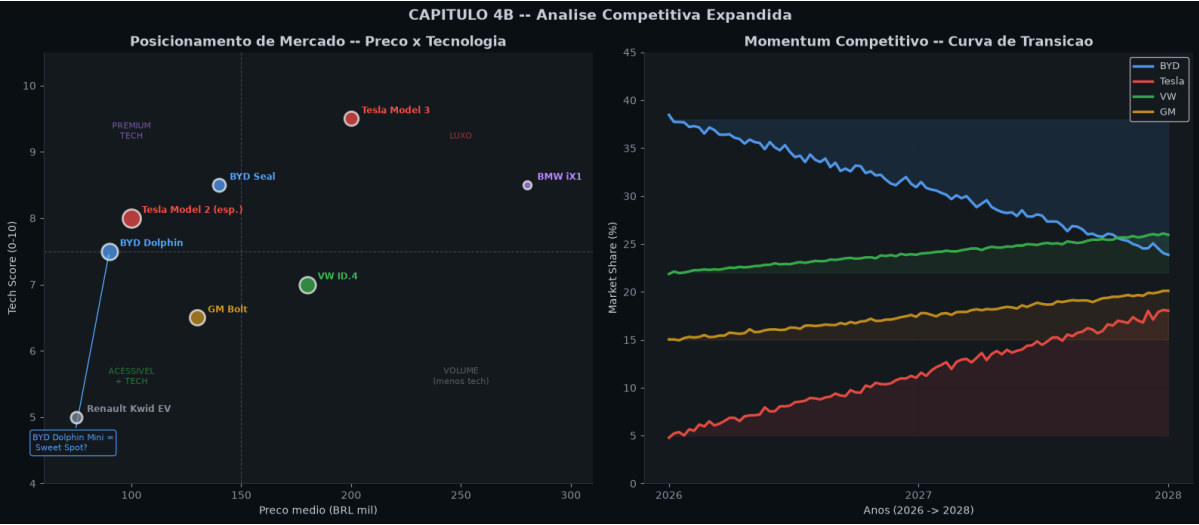


Figura salva: cap4b_competitive_deep.png

CAPITULO 5 — Índice Composto: O Raio-X da Vulnerabilidade

Vieses cognitivos fazem com que seres humanos deem peso desproporcional ao ultimo dado que viram. O indice composto existe para resolver isso: **uma nota unica que agrega todas as dimensoes de risco**, ponderadas por relevancia estrategica.

Dimensao	Peso	Score (0-100)	Status
Cambio	30%	70,9	Elevado
Supply Chain	30%	95,7	Critico
Regulatorio	20%	72,0	Elevado
Competitivo	20%	36,8	Moderado

Nota composta: 71,8/100 — Vulnerabilidade **ELEVADA**

```

In [11]: %matplotlib inline
from matplotlib.patches import Polygon, Circle

comp_data = D['composite']
scores = comp_data['scores']
weights = comp_data['weights']
total = comp_data['composite_pct']

fig = plt.figure(figsize=(18, 8))
gs = gridspec.GridSpec(1, 2, figure=fig, wspace=0.3)

ax1 = fig.add_subplot(gs[0], projection='polar')
dims = list(scores.keys())
# dims are already lowercase: cambio, supply, reg, comp
n_dims = len(dims)
angs = np.linspace(0, 2*np.pi, n_dims, endpoint=False)
score_vals = [scores[d] for d in dims]

angs_full = np.concatenate([angs, [angs[0]]])
scores_full = score_vals + [score_vals[0]]

ax1.fill(angs_full, scores_full, color=C_BLUE, alpha=0.25)
ax1.plot(angs_full, scores_full, color=C_BLUE, lw=2.5, marker='o', markersize=
10)

for r in [25, 50, 75, 100]:
    theta_circle = np.linspace(0, 2*np.pi, 100)
    ax1.plot(theta_circle, [r]*100, color=C_GRAY, lw=0.5, alpha=0.3)

dim_labels = ['CAMBIO\n(30%)', 'SUPPLY\nCHAIN\n(30%)', 'REGULA-\nTORIO\n(2
0%)', 'COMPETI-\nTIVO\n(20%)']
for angle, label, score in zip(np.concatenate([angs, [angs[0]]]),
                                dim_labels, scores_full):
    ax1.text(angle, score + 7, f'{score:.0f}', ha='center', fontsize=9,
             color=C_BLUE, fontweight='bold')

ax1.set_xticks(angs)
ax1.set_xticklabels(dim_labels, fontsize=9, color='white')
ax1.set_ylim(0, 105)
ax1.set_yticks([])
ax1.set_title('Indice Composto -- Radar de Vulnerabilidade', fontweight='bol
d', pad=20, fontsize=12)

ax2 = fig.add_subplot(gs[1])
dims_w = list(weights.keys())
w_labels = ['Cambio\n30%', 'Supply Chain\n30%', 'Regulatorio\n20%', 'Competiti
vo\n20%']
w_vals = [weights[d] * 100 for d in dims_w]
w_scores = [scores[d] for d in dims_w]
w_contrib = [w * s / 100 for w, s in zip(w_vals, w_scores)]

running = [0]
for c in w_contrib:
    running.append(running[-1] + c)

x_pos = np.arange(len(dims_w))
for i, (x, start, end, label, sc) in enumerate(zip(x_pos, running[:-1], runnin

```

```

='white')
    ax2.text(x, start + (end-start)/2 - 5,
             f'({w_vals[i]:.0f}%x{sc:.0f})',
             ha='center', va='top', fontsize=7.5, color='white', alpha=0.7)

ax2.set_xticks(x_pos)
ax2.set_xticklabels(w_labels, fontsize=9)
ax2.set_ylabel('Score Ponderado Acumulado', fontsize=10)
ax2.set_title(f'Decomposicao do Indice Composto\n(Total: {total:.1f}/100)', fontweight='bold')
ax2.set_ylim(0, 100)
ax2.axhline(total, color=C_YELLOW, lw=1.5, linestyle='--', label=f'Total: {total:.1f}/100')
ax2.grid(True, alpha=0.2, axis='y')
ax2.legend(fontsize=9)
ax2.text(x_pos[-1]+0.3, running[-1], f'{running[-1]:.1f}', va='center',
        fontsize=12, fontweight='bold', color=C_YELLOW)

fig.suptitle('CAPITULO 5 -- Indice Composto: O Raio-X da Vulnerabilidade',
            fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('cap5_composite.png', dpi=150, bbox_inches='tight', facecolor=fig.get_facecolor())
plt.show()
print('Figura salva: cap5_composite.png')

```

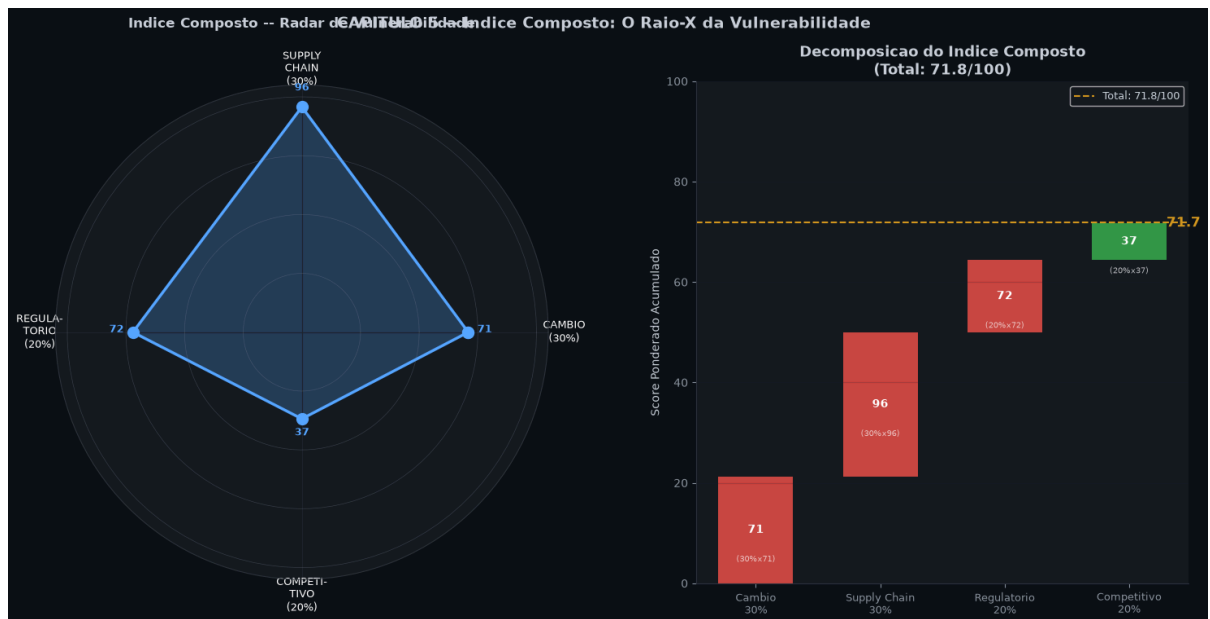


Figura salva: cap5_composite.png


```

In [12]: %matplotlib inline
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

ax = axes[0]
dims_s = ['Cambio', 'Supply Chain', 'Regulatorio', 'Competitivo']
score_keys = ['cambio', 'supply', 'reg', 'comp']
weights_list = [30, 30, 20, 20]
scores_list = [scores[k] for k in score_keys]
marginal_impact = [w * s / 100 for w, s in zip(weights_list, scores_list)]
colors_sens = [C_RED if s > 70 else C_YELLOW if s > 50 else C_GREEN for s in scores_list]
bars = ax.barh(dims_s, marginal_impact, color=colors_sens, alpha=0.8, height=0.5)
for bar, w, s, mi in zip(bars, weights_list, scores_list, marginal_impact):
    ax.text(mi + 0.3, bar.get_y() + bar.get_height()/2,
            f'{w}%x{s:.0f} = {mi:.1f}pp', va='center', fontsize=9,
            fontweight='bold', color=bar.get_facecolor())
ax.axvline(total, color=C_YELLOW, lw=2, linestyle='--', label=f'Indice total: {total:.1f}/100')
ax.set_xlabel('Contribuicao para o Indice (pp)', fontsize=10)
ax.set_title('Contribuicao Marginal por Dimensao\n(Peso x Score)', fontweight='bold')
ax.legend(fontsize=9); ax.grid(True, alpha=0.2, axis='x')

ax = axes[1]
improvement = 10
current_scores = [scores[k] for k in score_keys]
indices_after = []
for dim_idx in range(4):
    new_scores = current_scores.copy()
    new_scores[dim_idx] = max(0, new_scores[dim_idx] - improvement)
    new_total = sum(weights_list[i] * new_scores[i] / 100 for i in range(4))
    indices_after.append(new_total)

reduction = [total - ia for ia in indices_after]
bars2 = ax.barh(dims_s, reduction, color=colors_sens, alpha=0.8, height=0.5)
for bar, red in zip(bars2, reduction):
    ax.text(red + 0.1, bar.get_y() + bar.get_height()/2,
            f'-{red:.1f}pp', va='center', fontsize=10, fontweight='bold',
            color=bar.get_facecolor())
ax.set_xlabel('Reducao do Indice Composto (pp)', fontsize=10)
ax.set_title(f'Sensibilidade: "Se eu melhorar {improvement}pp..." \n(Indice atual: {total:.1f})', fontweight='bold')
ax.grid(True, alpha=0.2, axis='x')
ax.text(0.98, 0.02,
        f'Reduzir Supply Chain de 95,7 para 85,7\n'
        f'diminui o indice em 3,0pp (de {total:.1f} para {total-3:.1f})',
        transform=ax.transAxes, ha='right', va='bottom', fontsize=8.5, color=C_RED, alpha=0.9,
        bbox=dict(boxstyle='round', facecolor='#161b22', edgecolor=C_RED, alpha=0.8))

fig.suptitle('CAPITULO 5B -- Sensibilidade: Onde Intervir para Maximo Impacto?',
            fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('cap5b_sensitivity.png', dpi=150, bbox_inches='tight', facecolor=f

```



```

In [13]: %matplotlib inline
def score_color(s):
    if s >= 80: return C_RED
    if s >= 60: return C_ORANGE
    if s >= 40: return C_YELLOW
    return C_GREEN

def score_label(s):
    if s >= 80: return 'CRITICO'
    if s >= 60: return 'ELEVADO'
    if s >= 40: return 'MODERADO'
    return 'BAIXO'

fig = plt.figure(figsize=(20, 14))
gs = gridspec.GridSpec(3, 4, figure=fig, hspace=0.45, wspace=0.35)

# KW1: PTAX
ax = fig.add_subplot(gs[0, 0])
ax.text(0.5, 0.85, 'PTAX', fontsize=11, fontweight='bold', ha='center', transform=ax.transAxes)
ax.text(0.5, 0.65, f'BRL {D["ptax"]["atual"]}', fontsize=22, fontweight='bold',
        ha='center', color=C_BLUE, transform=ax.transAxes)
ax.text(0.5, 0.48, f'Vol: {D["ptax"]["vol30_annualized"]} a.a.',
        fontsize=9, ha='center', color=C_GRAY, transform=ax.transAxes)
ax.text(0.5, 0.35, f'Trend 180d: {D["ptax"]["trend180d_pct"]:+.2f}%',
        fontsize=9, ha='center', color=C_GRAY, transform=ax.transAxes)
ax.text(0.5, 0.18, f'Cambio Score: {scores["cambio"]:.0f}/100',
        fontsize=10, fontweight='bold', ha='center', color=score_color(scores['cambio']), transform=ax.transAxes)
ax.text(0.5, 0.07, score_label(scores['cambio']), fontsize=8, ha='center',
        color=score_color(scores['cambio']), transform=ax.transAxes, alpha=0.8)
ax.set_title('1. Cambio', fontweight='bold', fontsize=11, color=C_BLUE)
ax.axis('off')

# KW2: Supply HHI
ax = fig.add_subplot(gs[0, 1])
ax.text(0.5, 0.85, 'Supply HHI', fontsize=11, fontweight='bold', ha='center', transform=ax.transAxes)
ax.text(0.5, 0.65, f'{D["supply_hhi"]["battery_cells"]}', fontsize=22, fontweight='bold',
        ha='center', color=C_RED, transform=ax.transAxes)
ax.text(0.5, 0.48, 'Baterias (CATL 65%)', fontsize=9, ha='center', color=C_GRAY, transform=ax.transAxes)
ax.text(0.5, 0.35, f'Litio: {D["supply_hhi"]["lithium"]} | Semi: {D["supply_hhi"]["semiconductors"]}',
        fontsize=8, ha='center', color=C_GRAY, transform=ax.transAxes)
ax.text(0.5, 0.18, f'Supply Score: {scores["supply"]:.0f}/100',
        fontsize=10, fontweight='bold', ha='center', color=score_color(scores['supply']), transform=ax.transAxes)
ax.text(0.5, 0.07, score_label(scores['supply']), fontsize=8, ha='center',
        color=score_color(scores['supply']), transform=ax.transAxes, alpha=0.8)
ax.set_title('2. Supply Chain', fontweight='bold', fontsize=11, color=C_RED)
ax.axis('off')

```

```

# KW3: Regulatorio
ax = fig.add_subplot(gs[0, 2])
ax.text(0.5, 0.85, 'Regulatorio', fontsize=11, fontweight='bold', ha='center',
transform=ax.transAxes)
ax.text(0.5, 0.65, f'{D["reg_scenarios"]["continuidade"]:.0f}%', fontsize=22,
fontweight='bold',
        ha='center', color=C_BLUE, transform=ax.transAxes)
ax.text(0.5, 0.48, 'Incentivos (Rota 2030)', fontsize=9, ha='center', color=C_
GRAY, transform=ax.transAxes)
ax.text(0.5, 0.35, 'Rollback total: 0% | Expansao: 25%',
        fontsize=8, ha='center', color=C_GRAY, transform=ax.transAxes)
ax.text(0.5, 0.18, f'Reg Score: {scores["reg"]:.0f}/100',
        fontsize=10, fontweight='bold', ha='center', color=score_color(scores
['reg']), transform=ax.transAxes)
ax.text(0.5, 0.07, score_label(scores['reg']), fontsize=8, ha='center',
        color=score_color(scores['reg']), transform=ax.transAxes, alpha=0.8)
ax.set_title('3. Regulatorio', fontweight='bold', fontsize=11, color=C_YELLOW)
ax.axis('off')

# KW4: Competitivo
ax = fig.add_subplot(gs[0, 3])
ax.text(0.5, 0.85, 'Competitivo', fontsize=11, fontweight='bold', ha='center',
transform=ax.transAxes)
ax.text(0.5, 0.65, f'BYD {D["competition"]["byd"][0]*100:.0f}%', fontsize=22,
fontweight='bold',
        ha='center', color=C_BLUE, transform=ax.transAxes)
ax.text(0.5, 0.48, f'2026 share lider', fontsize=9, ha='center', color=C_GRAY,
transform=ax.transAxes)
ax.text(0.5, 0.35, f'Tesla: 5%->18% (ameaca 2027)',
        fontsize=8, ha='center', color=C_RED, transform=ax.transAxes)
ax.text(0.5, 0.18, f'Comp Score: {scores["comp"]:.0f}/100',
        fontsize=10, fontweight='bold', ha='center', color=score_color(scores
['comp']), transform=ax.transAxes)
ax.text(0.5, 0.07, score_label(scores['comp']), fontsize=8, ha='center',
        color=score_color(scores['comp']), transform=ax.transAxes, alpha=0.8)
ax.set_title('4. Competitivo', fontweight='bold', fontsize=11, color=C_GREEN)
ax.axis('off')

# KW5: Composite Gauge
ax = fig.add_subplot(gs[1, :2])
score = total
theta_gauge = np.linspace(np.pi, 0, 100)
r_gauge = 1
for th in np.linspace(np.pi, 0, 100):
    color_g = C_GREEN if th > 2*np.pi/3 else C_YELLOW if th > np.pi/3 else C_R
ED
    ax.plot([0, r_gauge * np.cos(th)], [0, r_gauge * np.sin(th)],
            color=color_g, lw=18, alpha=0.4)

needle_angle = np.pi * (1 - score/100)
ax.arrow(0, 0, 0.7 * np.cos(needle_angle), 0.7 * np.sin(needle_angle),
        head_width=0.08, head_length=0.05, fc='white', ec='white', lw=2)
ax.text(0, -0.25, f'{score:.1f}/100', fontsize=28, fontweight='bold',
        ha='center', va='center', color=score_color(score))
ax.text(0, -0.45, 'INDICE COMPOSTO', fontsize=10, ha='center', color=C_GRAY, s
tyle='italic')
ax.text(0, -0.58, score_label(score), fontsize=12, fontweight='bold', ha='cent

```

```

er', color=score_color(score))
ax.set_xlim(-1.3, 1.3); ax.set_ylim(-0.7, 1.1); ax.axis('off')
ax.set_title('5. Indice Composto -- Vulnerabilidade Geral', fontweight='bold',
fontsize=12, pad=10)

# KW6: Monte Carlo
ax = fig.add_subplot(gs[1, 2:])
bom_sample = np.array(D['bom_impacts_sample'])
p5_s, p50_s, p95_s = np.percentile(bom_sample, 5), np.percentile(bom_sample, 50), np.percentile(bom_sample, 95)
prob_loss_s = (bom_sample < 0).mean() * 100
ax.text(0.5, 0.92, 'Monte Carlo -- Impacto no BOM (10k x 6m)',
        fontsize=10, fontweight='bold', ha='center', transform=ax.transAxes)
kpi_data = [
    ('P5 (Risco)', f'{p5_s:.2f}pp', C_RED),
    ('P50 (Base)', f'{p50_s:.2f}pp', C_YELLOW),
    ('P95 (Oportun.)', f'{p95_s:.2f}pp', C_GREEN),
    ('Prob(perda)', f'{prob_loss_s:.1f}%', C_RED if prob_loss_s > 40 else C_YELLOW),
]
for i, (lbl, val, clr) in enumerate(kpi_data):
    x = 0.15 + i * 0.22
    ax.text(x, 0.65, lbl, fontsize=8, ha='center', transform=ax.transAxes, color=C_GRAY)
    ax.text(x, 0.42, val, fontsize=16, fontweight='bold', ha='center', transform=ax.transAxes, color=clr)
ax.set_xlim(0, 1); ax.set_ylim(0, 1); ax.axis('off')
ax.set_title('6. Simulacao Probabilistica', fontweight='bold', fontsize=12, pad=10)

# KW7: Timeline
ax = fig.add_subplot(gs[2, :])
risk_timeline = [
    {'t': '2026-Q3', 'event': 'Tesla Model 2\npre-venda Brasil', 'risk': 'COMP', 'level': 'MEDIUM'},
    {'t': '2026-Q4', 'event': 'Decisao Rota 2030\nrenovacao/rollback', 'risk': 'REG', 'level': 'HIGH'},
    {'t': '2027-Q1', 'event': 'Eleicoes presidenciais\n(impacto cambio)', 'risk': 'CAMBIO', 'level': 'HIGH'},
    {'t': '2027-Q2', 'event': 'Lancamento\nTesla Model 2', 'risk': 'COMP', 'level': 'CRITICAL'},
    {'t': '2027-Q3', 'event': 'Greve nacional\n(risco supply)', 'risk': 'SUPPLY', 'level': 'MEDIUM'},
    {'t': '2028-Q1', 'event': 'Expansao VW/GM\nline-up EV', 'risk': 'COMP', 'level': 'HIGH'},
]
level_colors = {'LOW': C_GREEN, 'MEDIUM': C_YELLOW, 'HIGH': C_ORANGE, 'CRITICAL': C_RED}
for i, ev in enumerate(risk_timeline):
    x = i / (len(risk_timeline) - 1)
    clr = level_colors[ev['level']]
    ax.scatter(x, 0.5, s=400, color=clr, alpha=0.8, zorder=5, edgecolors='white', linewidths=2)
    ax.text(x, 0.5, ev['t'].split('-')[1], ha='center', va='center',
            fontsize=7, fontweight='bold', color='white')
    ax.text(x, 0.22, ev['event'], ha='center', va='top', fontsize=7.5,
            color=C_GRAY, multialignment='center')

```

```
clr, alpha=0.8)
ax.axhline(0.5, color=C_GRAY, lw=1, alpha=0.3)
ax.set_xlim(-0.05, 1.05); ax.set_ylim(0, 1); ax.axis('off')
ax.set_title('Timeline de Riscos -- 2026 a 2028 (Eventos-Chave)', fontweight
='bold', fontsize=12, pad=10)

fig.text(0.5, 0.01,
        f'BYD Brasil -- Dashboard de Vulnerabilidade | Atualizado: {D["comp
uted_at"]}] | '
        f'PTAX: BRL {D["ptax"]["atual"]}] | Composite: {total:.1f}/100 | '
        f'Produzido por: Matheus Mendes',
        ha='center', fontsize=8, color=C_GRAY, style='italic')
fig.suptitle('DASHBOARD COMPLETO -- BYD Brasil / Camacari 2025-2027',
            fontsize=16, fontweight='bold', y=0.98)
plt.savefig('cap6_dashboard_final.png', dpi=150, bbox_inches='tight', facecolo
r=fig.get_facecolor())
plt.show()
print('Figura salva: cap6_dashboard_final.png')
```

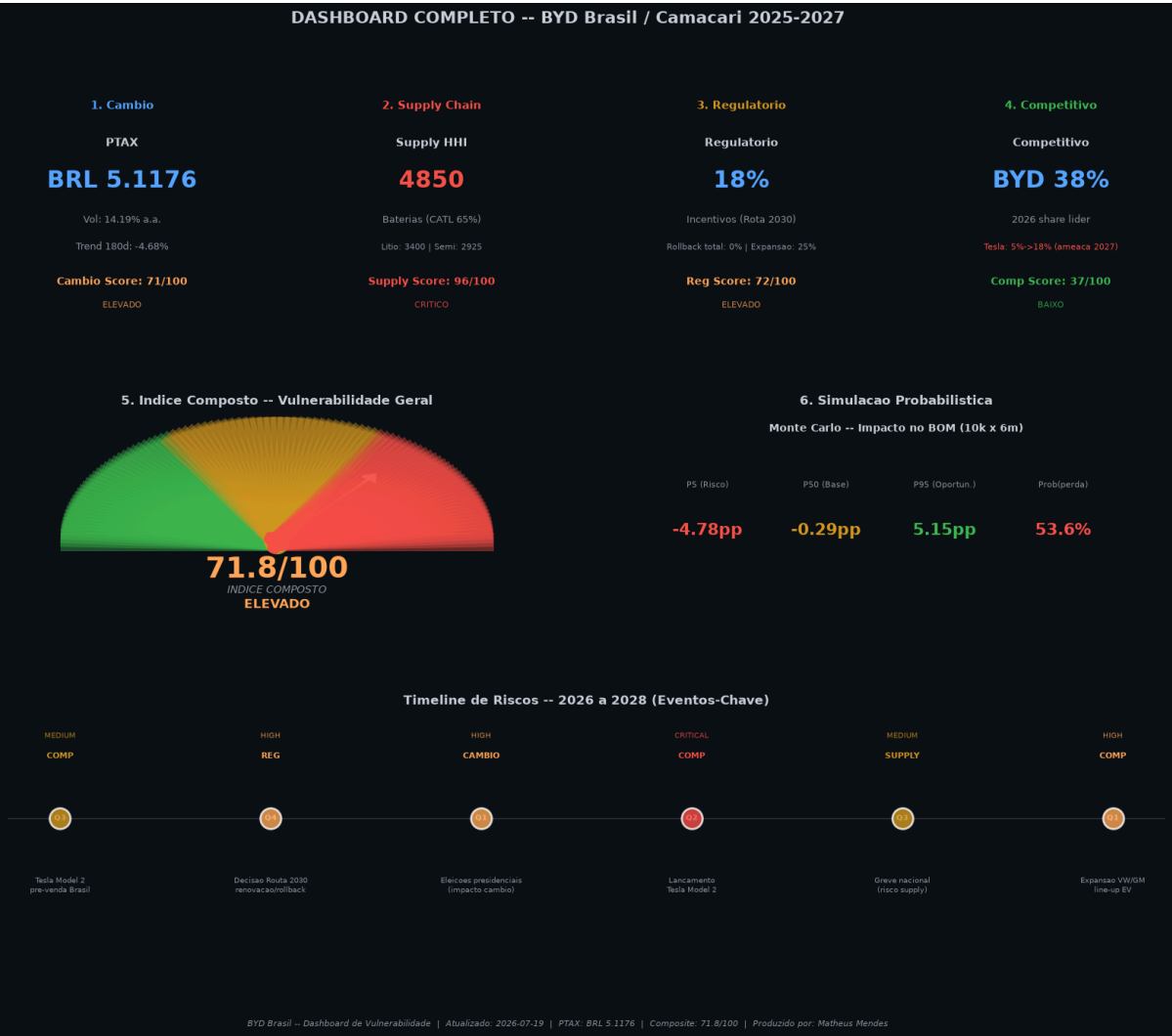


Figura salva: cap6_dashboard_final.png


```

In [14]: %matplotlib inline
print('=' * 70)
print('RESUMO EXECUTIVO -- BYD BRASIL / CAMACARI 2025-2027')
print('=' * 70)
print()
print(f"Data de referencia : {D['computed_at']}")
print(f"PTAX atual          : BRL {D['ptax']['atual']} (17/jul/2026)")
print(f"Volatilidade 30d     : {D['ptax']['vol30_annualized']}% a.a.")
print(f"Tendencia 180d       : {D['ptax']['trend180d_pct']:+.2f}% (depreciacao)")
print()
print('-' * 70)
print('INDICE COMPOSTO DE VULNERABILIDADE')
print('-' * 70)
for dim_key in ['cambio', 'supply', 'reg', 'comp']:
    label = {'cambio': 'Cambio', 'supply': 'Supply Chain', 'reg': 'Regulatori
o', 'comp': 'Competitivo'}[dim_key]
    print(f" {label:15s} : {scores[dim_key]:5.1f}/100 [{score_label(scores[d
im_key])}]")
print(f" {'-'*15} : {'-'*10}")
print(f" {'COMPOSTO TOTAL':15s} : {total:5.1f}/100 [{score_label(total)}]")
print()
print('-' * 70)
print('MONTE CARLO -- 10.000 SIMULACOES x 6 MESES')
print('-' * 70)
print(f" P5 (risco severo, 90% conf) : {p5:.2f}pp")
print(f" P50 (mediana/base)           : {p50:.2f}pp")
print(f" P95 (otimismo)                : {p95:.2f}pp")
print(f" Probabilidade de perda         : {prob_loss:.1f}%")
print()
print('-' * 70)
print('STRESS TEST -- APRECIACAO DO REAL')
print('-' * 70)
for s in D['stress']:
    print(f" {s:>5} PTAX BRL {D['stress'][s]['ptax']:.4f} -> BOM {D['stres
s'][s]['bom']:+.1f}pp")
print()
print('-' * 70)
print('SUPPLY CHAIN -- HHI')
print('-' * 70)
print(f" Celulas de Bateria : HHI {D['supply_hhi']['battery_cells']} (CATL 6
5%)")
print(f" Litio              : HHI {D['supply_hhi']['lithium']}")
print(f" Semicondutores      : HHI {D['supply_hhi']['semiconductors']} (TSMC 4
0%)")
print()
print('-' * 70)
print('MARKET SHARE -- PROJECAO 2026-2028')
print('-' * 70)
for brand in ['byd', 'tesla', 'vw', 'gm', 'others']:
    vals = comp[brand]
    delta = (vals[-1] - vals[0]) * 100
    print(f" {brand.upper():8s} {vals[0]*100:5.1f}% (2026) -> {vals[-1]*100:
5.1f}% (2028) d={delta:+.1f}pp")
print()
print('=' * 70)
print('PRIORIDADES DE ACAO')
print('=' * 70)

```


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RESUMO EXECUTIVO -- BYD BRASIL / CAMACARI 2025-2027

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Data de referencia : 2026-07-19
PTAX atual : BRL 5.1176 (17/jul/2026)
Volatilidade 30d : 14.19% a.a.
Tendencia 180d : -4.68% (depreciacao)

INDICE COMPOSTO DE VULNERABILIDADE

Cambio	:	70.9/100	[ELEVADO]
Supply Chain	:	95.7/100	[CRITICO]
Regulatorio	:	72.0/100	[ELEVADO]
Competitivo	:	36.8/100	[BAIXO]
-----	:	-----	
COMPOSTO TOTAL	:	71.8/100	[ELEVADO]

MONTE CARLO -- 10.000 SIMULACOES x 6 MESES

P5 (risco severo, 90% conf)	:	-4.78pp
P50 (mediana/base)	:	-0.29pp
P95 (otimismo)	:	5.15pp
Probabilidade de perda	:	53.6%

STRESS TEST -- APRECIACAO DO REAL

Base PTAX BRL 5.1176	->	BOM +0.0pp
-5% PTAX BRL 4.8617	->	BOM -2.1pp
-10% PTAX BRL 4.6058	->	BOM -4.2pp
-20% PTAX BRL 4.0941	->	BOM -8.4pp
-30% PTAX BRL 3.5823	->	BOM -12.6pp

SUPPLY CHAIN -- HHI

Celulas de Bateria	:	HHI 4850 (CATL 65%)
Litio	:	HHI 3400
Semicondutores	:	HHI 2925 (TSMC 40%)

MARKET SHARE -- PROJECAO 2026-2028

BYD	38.0% (2026)	->	24.0% (2028)	d=-14.0pp
TESLA	5.0% (2026)	->	18.0% (2028)	d=+13.0pp
VW	22.0% (2026)	->	26.0% (2028)	d=+4.0pp
GM	15.0% (2026)	->	20.0% (2028)	d=+5.0pp
OTHERS	20.0% (2026)	->	12.0% (2028)	d=-8.0pp

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PRIORIDADES DE ACAO

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1. [CRITICA] Qualificar fornecedor #2 de baterias (Samsung SDI ou LG)
2. [ALTA] Implementar hedge cambial NDF 6m (50% da exposicao)

